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MILITARY SPECIFICATION

ACCELEROMETER, AIRCRAFT MX-6663/AJB-7

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of aircraft accelerometer, designated MX-6663/AJB-7.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-packaging, Methods Of
DOD-D-1000	Drawings, Engineering and Associated Lists
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-C-38295	Computer Set, Attitude - Reference Bombing AN/AJB-7

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-470	Maintainability Program Requirements (For Systems And Equipment)
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the accelerometer except when magnetic materials are essential.

3.3.2 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against deterioration in a manner that will in no way prevent

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compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided. Components contained within hermetic seal shall be considered to be adequately protected against deterioration.

3.3.3 Fungus-proof. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if used in a hermetically sealed enclosure, fungicidal treatment will not be required.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met during storage or normal service.

3.3.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.6 Corrosive fumes. Materials as installed in the accelerometer and under the service conditions specified herein shall not liberate deleterious fumes.

3.3.7 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

3.4 Design and construction.

3.4.1 General design. The accelerometer specified herein is a component of the AN/AJB-7 computer set conforming to MIL-C-38295, and shall be designed to sense aircraft acceleration normal to the logitudinal axis of the aircraft.

3.4.1.1 Warmup period. The accelerometer shall not require a warmup period prior to meeting the operational requirements specified herein. The accelerometer shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1.2 Adjustments and repairs. The accelerometer shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases.

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3.4.2 Compatibility. The accelerometer shall meet all applicable requirements of the AN/AJB-7 computer set conforming to MIL-C-38295.

3.4.3 Maintainability. The following maintainability requirements shall be considered in the design of the accelerometer:

a. Minimization of the complexity of maintenance tasks (for example, calibration, adjustments, inspections, etc.) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and postive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly, or component

d. Minimum number and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal, or replacement.

3.4.4 Reliability. The accelerometer shall be so designed that when tested as specified in 4.6.25, the Mean-Time-Between-Failure (MTBF) (θ1) shall be at least 2.0 and confidence level will be 70%. Test Plan VIIIC shall be used as specified in MIL-STD-781. MIL-STD-785 shall be used as a guide for proper implementation of the required reliability into the design and production of the equipment.

3.4.4.1 Longevity. The accelerometer shall have an operating life span of not less than 2,000 hours before the equipment consistently fails to meet the MTBF index.

3.4.5 Construction. The accelerometer shall consist of a spring-supported mass and a precision potentiometer electrical pickoff device enclosed in a sealed container.

3.4.6 Input acceleration. The accelerometer shall be designed to operate for linear accelerations up to and including 5g from a 0g position. Mechanical stops shall be set so that no electrical discontinuities will be obtained from steady-state accelerations up to 10g.

3.4.7 Output voltage. The output voltage shall be proportional to the linear input acceleration along the sensitive or input axis.

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3.4.8 Pickoff. The potentiometer-type pickoff shall have a resistance of 2,000 ohms, +10, -0 percent and shall be padded on each end with an additional resistance of 500 to 515 ohms, equally matched within 1 percent with a temperature coefficient equal to the potentiometer within 10 percent. The maximum resistance variation through the applicable temperature range shall be ±6 percent.

3.4.9 Calibration and mounting. The accelerometer shall be calibrated with its base down. When mounted with its base up, the calibrated voltage ratio (output to input) shall not vary by more than 0.01 at the 0.0g offset nor by more than 0.03 at any other g-value within the range of the instrument. The accelerometer shall be operable within the limits of calibration when mounted with its sensitive axis vertical.

3.5 Performance. The accelerometer shall be capable of satisfactory operation after subjection to any combination of the following conditions:

a. Temperatures - temperatures ranging from -62° to +85°C

b. Relative humidity - up to 100 percent

c. Vibration - An applied double amplitude of 0.10 inch from 5 to 14 cps, 1g from 14 to 23 cps, 0.036 inch double amplitude from 23 to 52 cps, and 5g from 52 to 500 cps

d. Acceleration - acceleration forces up to 20g

e. Altitude - pressures ranging from 30 inches Hg (sea level) to 0.082 inch Hg (approximately 80,000 feet)

f. Fungus - exposure to fungus enzymes

g. Salt fog - exposure to salt fog for a period of 50 hours

h. Sand and dust - exposure to blowing fine sand and dust particles

i. Temperature shock - abrupt temperature change from +85° to -40°C

j. Dynamic shock - forces of 15g operating and 30g nonoperating for 11 milliseconds

k. Acoustical noise - exposure to acoustic sound fields that are characteristic of aircraft

3.5.1 Radio noise interference. The accelerometer shall meet the radio noise interference requirements of MIL-I-6181.

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3.5.2 Dielectric strength. The accelerometer shall withstand a potential of 500V rms at 50 to 60 cps applied for 1 minute between terminals A, B, and C (connected together) of the connector and the accelerometer case.

3.5.3 Power. The accelerometer shall operate with 30 ±0.5V dc for pickoff excitation. The pickoff shall be capable of carrying a continuous duty current of 15 milliamperes. This power shall be supplied by the AN/AJB-7 system.

3.5.4 Natural frequency. The accelerometer shall have a natural resonant frequency of 11.0 cps ±10 percent.

3.5.5 Damping. The accelerometer shall have a damping ratio of 0.34 to 1.5 over the temperature range of -54° to +71°C.

3.5.6 Resolution. The average increment of voltage ratio (output voltage to input voltage) within the range of the instrument shall be 0.004 maximum.

3.5.7 Static friction. The maximum acceleration required to overcome friction under static conditions (no vibration) shall be 0.03g.

3.5.8 Hysteresis. The maximum difference between the voltage ratio at any g-value during increasing acceleration and the voltage ratio at the same g-value during decreasing acceleration shall be 0.01.

3.5.9 Sensitivity. The sensitivity shall be as follows:

Acceleration	Voltage Ratio with Padding Resistors
0.0	0.138 to 0.198
0.5	0.204 to 0.264
1.0	0.271 to 0.331
1.5	0.337 to 0.397
2.0	0.404 to 0.464
2.5	0.470 to 0.530
3.0	0.536 to 0.596
3.5	0.602 to 0.662
4.0	0.669 to 0.729
4.5	0.735 to 0.795
5.0	0.802 to 0.862

3.5.10 Cross coupling. The output signal variation with acceleration perpendicular to the sensitive axis shall be less than 3 percent of the value realized if the acceleration is applied to the sensitive axis.

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3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identificaiton and part number requirements of DOD-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.7 Connector. The connector shall be a Bendix PT-02H-10-6P connector, or equivalent.

3.8 Pin utilization. Pin utilization shall be as follows:

Pin	Use
A	−
C	+ } 30V d-c input
B	Output with respect to A
D	Spare
E-F	Spare

3.9 Case. The case shall conform to figure 1. The case shall be entirely convered with a durable, dull black finish. The sensitive axis and the direction of sensitivity shall be suitably marked on the case.

3.9.1 Sealing. The case shall provide a hermetically sealed inclosure for all mechanical and electrical components of the accelerometer. The design of the case shall be such that the internal mechanism may be removed from the case as a unit, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the contracting activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the transmitter may be subjected.

3.9.2 Filling medium. The filling medium shall be at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. Where practical, the 100 percent helium filling medium shall be utilized. If the accelerometer is a floated unit and is sealed in a compartment of the case separate from the other components, this requirement shall apply only to the compartment containing the remaining components.

3.10 Shockmount. A shockmount shall not be provided between the accelerometer and the airframe.

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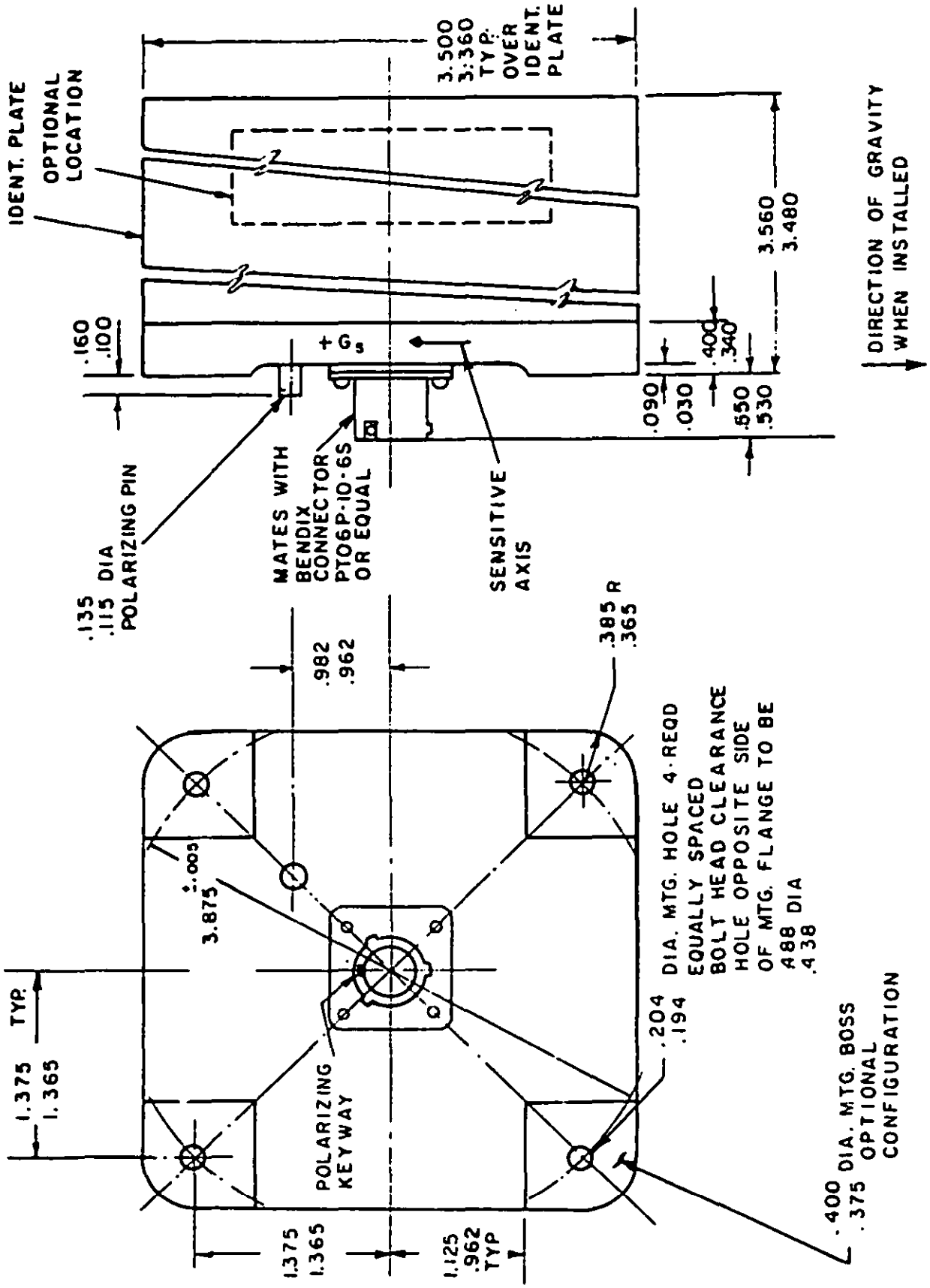


FIGURE 1. Case

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3.11 Weight. The weight of the accelerometer, excluding cables, shall not exceed 3 pounds.

3.12. Dimensions. The dimensions shall conform to figure 1.

3.13 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.14 Installation.

3.14.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the accelerometer. They shall have a durable, dull black, oxidized or black nickel finish.

3.14.2 Envelope. An envelope containing mounting screws shall be furnished by the contractor and shall be packed with each instrument. The envelope shall be marked with the following information:

IMPORTANT

THIS ENVELOPE CONTAINS

MOUNTING SCREWS

3.15 Finishes and protective coatings.

3.15.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Small holes and case inserts need not be anodized.

b. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 shall be used in lieu of anodizing.

d. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

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3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The sensitive axis and direction of sensitivity shall be clearly marked on the case.

3.17 Workmanship. The accelerometer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges as well as resin flash that may crumble.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the accelerometer, they shall be held or limited accordingly.

3.17.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.17.3 Cleaning. The accelerometer shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign materials removed after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirement as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Goverment. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of accelerometers shall be classified as follows:

a. First article tests

b. Acceptance tests

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test will be made

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at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with the atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Voltage input. Unless otherwise specified, the accelerometer shall be tested with the input stated in 3.5.3.

4.3.3 Attitude. Unless otherwise specified, the accelerometer shall be tested in its normal operating position.

4.3.4 Friction remover. An Edwards vibrator, or equivalent, shall be used to remove friction before a test reading is taken. Tapping the instrument will not be acceptable. Friction within the instrument shall be sufficiently low that it can be removed by the vibrator. If the vibrator is used for any one test, it shall be used throughout all testing specified under 4.6.

4.3.5 Test sequence. Unless otherwise specified, the tests shall be conducted in the sequence listed under 4.6.

4.4 First article testing.

4.4.1 Test samples. The test samples shall consist of accelerometers representative of the production equipment. Three of the accelerometers shall be subject to all the tests except reliability. The test sample size for the reliability test shall be of sufficient size to accumulate the required number of test hours for verification of the test requirements in a minimum number of days as specified by contract, or if not specified, not to exceed 90 days.

4.4.1.1 Reliability demonstration. Unless otherwise specified, the number of test samples for reliability testing will be selected at random from the first production on the contract or order and tested as specified in 4.6.25. Use Test Plan VIIIC, MIL-STD-781.

4.4.2 Test reports, samples, and data for the contracting activity. When tests are conducted at a location other than the laboratory of the contracting activity, the following shall be furnished to that activity:

a. Test report - three copies of a test report in accordance with MIL-STD-831

b. The samples that were tested

c. Two sets of assembly drawings

d. Brief operating instructions

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4.4.3 First article tests. The first article tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling plans and tests

4.5.1 Individual tests. Each accelerometer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Stabilization

c. Sensitivity

d. Friction and resolution

e. Output sensitivity

f. Hysteresis

g. Cross coupling

h. Natural frequency and damping

i. Sealing

j. Dielectric strength

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. One accelerometer shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Low temperature

c. High temperature

d. Vibration error

e. Voltage variation

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4.5.2.2 Sampling plan B. Unless otherwise specified, 3 accelerometers shall be selected at random from the first 15 items on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Radio noise interference

c. High temperature exposure

d. Accoustical noise

e. Altitude-temperature

f. Temperature shock

g. Dynamic shock

h. Vibration failure

i. Acceleration

j. Humidity

k. Salt spray

l. Sand and dust

m. Fungus

n. Sealing thermal shock

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests continuation. For operational reasons, individual tests may be continued pending the ivestigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.5.2.4 Empty cases. At the option of the contracting activity, the humidity, salt fog, sand and dust, fungus, and thermal shock tests may be conducted on empty cases with all external parts attached as in production units.

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4.5.3 Sampling plan C. Unless otherwise specified, 13 accelerometers shall be selected at random from each 50 or less produced on the contract or order and tested as specified in 4.6.25 for a minimum of 25,800 hours.

4.5.3.1 Acceptance or rejection. Test Plan VIIIC, MIL-STD-781 shall be the criteria used to determine whether the equipment will be accepted or rejected. Test records will be maintained per App D, MIL-STD-781. Environmental cycle will be per paragraph 4.6.25.

4.5.3.2 Recording. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.5.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The accelerometer shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking and configuration as shown on figure 1.

4.6.2 Stabilization. The accelerometer shall be subjectd to a 10-hour stabilization period. If a failure occurs, the unit shall be repaired and the entire stabilization procedure repeated. Each unit shall be subjected to all remaining tests of this procedure after completion of stabilization. The environment for the stabilization test shall be as follows:

Vibration:	0.02 inch double amplitude at 25-36 cps applied normal to the mounting plan for 100 minutes with power on either before or after the remaining 500 minutes.
On-off cycling:	On 4 hours and off 1 hour of each 5-hour period of test
D-C voltage:	30 ±0.5V

4.6.2.1 Cycling. During stabilization, 30 ±0.5V dc shall be applied between connector terminals A and C and a 3,000 ohm, ±10 percent, load shall be connected between terminals B and C. The acceleration applied along the sensitive axis shall be cycled from 0 to ±1g and to -1g at least 6 times during stabilization.

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4.6.2.2 Operational check. At the end of 5 hours of stabilization, the accelerometer shall be subjected to an operational check consisting of the test specified in 4.6.3 at 0.0g, +0.5g, and +1.0g only.

4.6.3 Sensitivity. When the accelerations specified below are applied, the voltage ratio shall be as follows:

Acceleration (g)	Voltage Ratio
0.0	0.138 to 0.198
+0.5	0.204 to 0.264
+1.0	0.271 to 0.331
+1.5	0.337 to 0.397
+2.0	0.404 to 0.464
+2.5	0.470 to 0.530
+3.0	0.536 to 0.596
+3.5	0.602 to 0.662
+4.0	0.669 to 0.729
+4.5	0.735 to 0.795
+5.0	0.802 to 0.862

Notes: Zero acceleration - zero acceleration is when the unit is at rest with the plane of the mounting surface perpendicular to the pull of gravity.

Voltage ratio - voltage ratio is the ratio of output voltage to input voltage. Output is read across pins A and B.

4.6.4 Friction and resolution. The accelerometer shall be adjusted for 0g and vibrated lightly. the vibration shall be stopped and the acceleration shall be changed slowly until a change in output is detected. The change in acceleration required to cause a change in output shall be not greater than 0.175g. The above shall be repeated at 0.2g.

4.6.5 Output sensitivity. The accelerometer shall be adjusted for 0g and vibrated lightly. The vibration shall be stopped and the acceleration slowly changed by 0.0375g. There shall be a detectable change in output if the accelerometer is then vibrated along either or both of the nonsensitive axes.

4.6.6 Hysteresis. Voltage ratios for accelerations of 0.0g, negative stop, 0.0g, 1.0g, 0.0g shall be obtained in the above order being careful not to overshoot the g-value. The difference in voltage ratio for the same g-value shall be not greater than 0.01.

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4.6.7 Cross coupling. An acceleration of 1.0g shall be applied to the sensitive axis and the output voltage between pins A (-) and B (+) shall be noted. An acceleration of 1.0g shall be applied to the insensitive axes (each one in turn) and the output voltage between pins A (-) and B (+) shall be noted. The difference between the insensitive axes readings and the null shall be less than 3 percent of the 1g sensitive axis output.

4.6.8 Natural frequency and damping. The accelerometer shall be mounted to a load-compensated vibration machine and power shall be applied as specified in 4.3.2. The accelerometer shall be vibrated with an increasing frequency until the natural frequency of resonance is reached; this frequency shall be a minimum of 8 cps.

4.6.8.1 Phase angle. A sinusoidal displacement of one-half natural frequency (determined by vibrating with an increasing frequency until the natural frequency at resonance is reached) and a magnitude of ±0.5g shall be applied. The phase angle between the input displacement and the output signal shall be 136° ±20°.

4.6.9 Sealing. The accelerometer shall be immersed in glycerine, diethylene glycol, or a saturated solution of sodium chloride in tap water. The supply liquid temperature shall be between 11° and 25°C (52° to 77°F). The absolute pressure of the air above the liquid shall then be reduced to approximately 1 inch of mercury and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is the longer. The absolute pressure shall then be increased to 2-1/2 inches of mercury. Any bubbles coming from within the accelerometer shall be considered as leakage and the unit rejected. Bubbles that are the result of entrapped air on the various, exterior parts shall not be considered as leaks.

4.6.10 Dielectric strength. A potential of 500V rms at 50 to 60 cps shall be applied for 1 minute between terminals A, B, and C (connected together) of the connector and the case of the accelerometer. There shall be no evidence of breakdown.

4.6.11 Low temperature. The accelerometer shall be subjected to a temperature of -54° ±2°C without power applied for a period of 4 hours. Normal power shall then be applied. The accelerometer shall meet the individual tests while at -54°C.

4.6.12 High temperature. Each sample shall be subjected to a temperature of 71° ±2°C with normal power applied for a period of 4 hours. The accelerometer shall meet the individual tests while at 71°C.

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4.6.13 Vibration error. The accelerometer shall be subjected to 0.05 g/cps ±10 percent throughout a frequency spectrum of 20 to 2,000 cps for 5 minutes in each of the units 3 planes. Normal power shall be applied and there shall be no discontinuities between pins A and B during vibration.

4.6.14 Voltage variation. The accelerometer shall meet the individual tests when the input is varied from 29 to 31V dc.

4.6.15 Radio noise interference. The accelerometer shall be subjected to the radio noise interference test for devices of this type specified in MIL-I-6181.

4.6.16 High temperature exposure. The high temperature test shall be conducted in accordance with method 501.2, procedure I of MIL-STD-810, except the temperature shall be 85°C for a period of 24 hours. While still at the high temperature, rated power shall be applied. The accelerometer shall operate without sticking. After the accelerometer has returned to room temperature, it shall pass the individual tests. There shall be no damage that would affect subsequent operation.

4.6.17 Acoustical noise. The accelerometer shall be subjected to an acoustical noise test in accordance with MIL-STD-810, method 515.2, procedure I, category A.

4.6.18 Tempercature-altitude. The accelerometer shall be subjected to a temperature-altitude test in accordance with MIL-STD-810, method 504, class 2, except that the altitude shall be 80,000 feet, step 6 shall be conducted at 85°C for 4 hours, and step 8 shall be conducted at 85°C.

4.6.19 Temperature shock. The temperature shock test shall be conducted in accordance with method 503.1, procedure I of MIL-STD-810. After the accelerometer has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The accelerometer shall be subjected to and shall pass the individual tests.

4.6.20 Dynamic shock. The dynamic shock test shall be conducted in accordance with method 516.2 procedures I and IV of MIL-STD-810. Following this test, the accelerometer shall be subjected to and shall meet the individual tests.

4.6.21 Vibration failure. The vibration failure test shall be conducted in accordance with method 514.2 equipment categroy b.1, procedure I, Table 514.2-II, figure 514.22, curves B and C of MIL-STD-810, with power applied. At the end of this test, the accelerometer shall be subjected to and shall pass the individual tests. No damage or irregular operation shall result.

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4.6.22 Acceleration. The accelerometer, not operating, shall be mounted on a centrifuge in its normal manner and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The accelerometer shall be subjected to an acceleration of 20g in each of the above axes for a period of 1 minute. At the end of this acceleration, the accelerometer shall be subjected to the individual tests. No damage to the accelerometer shall result from this test.

4.6.23 Environmental tests. The accelerometer shall be subjected to the following environmental tests in accordance with the specified procedures of MIL-STD-810. At the option of the contracting activity, these tests may be performed on finished empty cases. At the completion of each test, if empty cases are used, the case shall be inspected for any damage that would impair operation.

4.6.23.1 Humidity. The humidity test shall be conducted in accordance with procedure I, method 507.1.

4.6.23.2 Fungus. The fungus test shall be conducted in accordance with procedure I, method 508.1.

4.6.23.3 Salt fog. The salt fog test shall be conducted in accordance with procedure I, method 509.1.

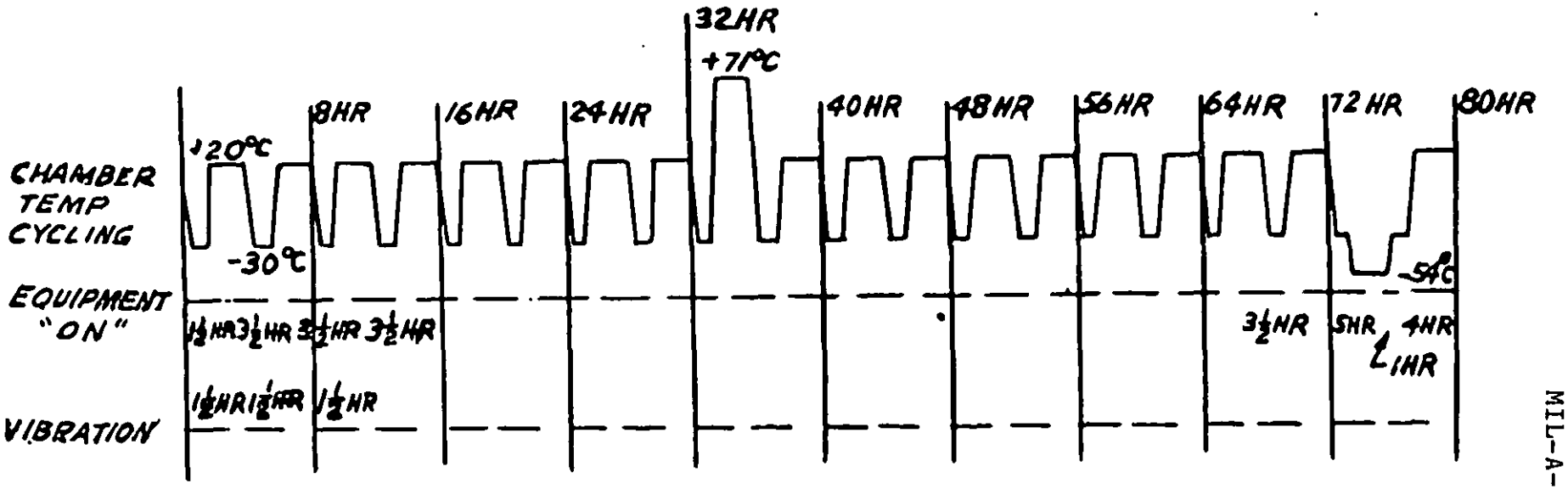
4.6.23.4 Sand and dust. The sand and dust test shall be conducted in accordance with procedure I, method 510.1.

4.6.24 Sealing thermal shock. the accelerometer, with mechanisms removed and sealed in the same manner as the complete accelerometer, shall be immersed alternately in tap water maintained at 85° ±5°C and at 5° ±4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes with not more than 5 seconds elapsing between bath immersions. No damage to the hermetic seal shall occur as a result of this test. Following the immersions, the accelerometer case shall be subjected to and shall meet the test specified in 4.6.9.

4.6.25 Reliability. The accelerometer shall be properly connected, with power supplied as specified in 3.5.3, and subjected to the reliability environmental cycles shown on figure 2. After each 160 hours of operation, the accelerometer shall meet the tests specified in 4.6.3 through 4.6.8.1.

4.6.25.1 Longevity. At the completion of the test specified in 4.6.25.3 of test accelerometers shall continue to be tested through a longevity test for a total of 2,000 hours on each computer.

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NOTES:

a. Environmental cycle repeats every 80 hours.

b. Temp rate of change of chamber shall be 0.1°C - 0.5°C per second.

c. Vibration level 0.2G at 20 to 30 CPS.

d. "Off" periods for equipment "on" and "off" are 1/2 hour (except 76th hour).

e. Monitoring of system outputs for catastrophic-type failures performed during each "on" cycle.

f. "Satisfactory performance" tests on systems and individual units conducted every 160 hours of environment cycle to detect performance deviation type failures.

FIGURE 2. Reliability test environment cycles.

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4.6.25.2 Relevancy criteria. All failures shall be classified as relevant failures unless approved for exclusion by the contracting activity.

4.6.25.3 Test information. Sixty days prior to the start of testing, the contractor shall submit detailed test information for contracting activity approval. This information shall include, but not necessarily be limited to:

a. Equipment used in monitoring performance

b. Periodicity of calibration checks to be made on monitoring equipment.

c. Description of environmental test facilities to be used, including recording facilities, method of control of heading and cooling facilities, and vibration equipment.

d. Frequency of monitoring

4.6.25.4 Test records. Throughout this test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for first article. As a minimum, the records shall include the following in accordance with MIL-STD-781: An operating sheet, log of failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the data shall be submitted to the contracting activity within 30 days after the completion of each test phase.

4.6.25.5 Repair of failed units. In the event of a failure during testing, the failed unit shall be repaired and the test continued.

4.6.26 Maintainability verification. Maintainability verification sall be in accordance with MIL-STD-470. All tasks selected shall be effectively simulated.

4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First article. When the unit container is capable of serving as the shipping container, first article inspection, and

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rough handling tests as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified by the contracting activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method 1C-1 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.7.2 of this specification.

5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, Table II of MIL-STD-794 applies.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather-resistant.

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5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The accelerometer covered by this specification is intended for use in the F-4C aircraft and is a component of the AN/AJB-7 computer set.

6.2 Ordering date. Contracting documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted

c. When sampling plan C tests will not be conducted

d. Levels of packaging and packing required.

6.2.1 Failure report. During testing, a report of each failure that occurs should be submitted to the contracting activity within 2 working days after the failure.

6.3 Definitions.

6.3.1 Hermetic seal. A hermetic seal is a perfectly closed and airtight seal made between vitric or metallic, or both, materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.2 Failure. Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition that fails to comply with the requirements specified herein.

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